

Color Sorting Machine

Learning Outcomes

By the end of the theme student will be able to:

- Select Optimal Cam Type.
- Identify the best belt type for your system.
- Research and apply methods to prevent jams.
- Ensure smooth object flow through the system.
- Identify Belt Drive Components: Recognize pulleys, shafts, and belts.
- Create conveyors and bins for sorting.
- Assemble System: Accurately build the sorting system.
- Create Linkages: Design linkages to control sorting arms or gates.
- Wire Color Sensor: Connect a color sensor to an electrical circuit.
- Control Multiple Servos: Connect and synchronize multiple servo motors.
- Calibrate Sensor: Test and fine-tune the color sensor.
- Power and Sync Servos: Configure the circuit to power and sync servos.
- Learn New Functions: Understand and apply new functions and loops.
- Link Signals: Connect sensor input to servo motor output.
- Integrate Arduino with Python: Interface Arduino with Python for control.

Contents

Lesson 1	3
1. Engage:	4
2. Think:	6
3. Explore:	7
4. Explain:	9
5. Elaborate:	13
6. Evaluate:	14

Lesson 1

UNIT 01: [Beginner]

Chapter 01: Mechanics

Lesson 01: Case scenario

Time Frame: 90 minutes

Objectives:

- Understand the importance of automation in sorting processes.
- Identify key components of an automated color sorting machine.
- Compare manual vs. automated sorting and analyze efficiency.
- Explore real-world applications of color sorting in industries.
- Recognize the role of sensors and motors in automation.

Challenge Questions of the Lesson

4. What are the main challenges in manual sorting?
5. How can automation improve sorting efficiency?
6. What industries use color sorting machines?

ESL Relation

Vocabulary Development:

Introduce key terms like:
Automation , Sorting System

Answers to Challenge Questions

1. Manual sorting is slow, inconsistent, and prone to human errors. It also requires more labor and time.
2. Automation speeds up sorting, improves accuracy, reduces labor costs, and ensures consistency in production.
3. Industries like food processing (grains, nuts), recycling (plastic sorting), and manufacturing (quality control in product assembly).

Answers of Exercises

- 1.

Activity Name	Time Allocated	Description
Engage: Case Scenario Exploration	10 min	Introduce a real-world factory problem, discuss inefficiencies in manual sorting.
Think: Role of Production Engineers	10 min	Discussion on how engineers improve production efficiency using automation.
Think: Brainstorm Web on Sorting System	10 min	Students identify components (sensors, motors) needed for an automated system.
Explore: Manual Sorting Activity	10 min	Hands-on sorting of colored balls into containers, reflecting on challenges.
Explore: Online Sorting Game	10 min	Students play the Nuts Puzzle game and analyze its sorting strategies.
Explain: Video on Production Engineers	10 min	Watch and discuss the role of production engineers in manufacturing.
Explain: Guided Reading on Automation	10 min	Students read about factory automation and discuss key takeaways.
Elaborate: Class Discussion & Reflection	10 min	Students summarize what they learned and relate it to the sorting machine project.
Evaluate: Quick Exit Ticket Reflection	10 min	Students write one key takeaway and one question they still have.

1. Engage:

Teacher Role:

- Facilitate discussion by presenting the case scenario.
- Ask thought-provoking questions to guide students in understanding the problem.
- Encourage students to think critically and relate the problem to real-world applications.
- Lead a hands-on mini-activity to demonstrate the need for automation.

Expected From Students:

- Actively listen to the case scenario.
- Participate in discussions by sharing their thoughts on why sorting is important in factories.
- Engage in a Think-Pair-Share activity to propose possible solutions.
- Take part in a hands-on sorting challenge to compare manual and automated processes.

Parts of the Books Included:

Project Overview

Adam and Laila's school organized a field trip to an industrial area where they visited one of the factories.

The candy factory has been experiencing a serious problem in the sorting department.

The color sorter machine, which is responsible for separating different colored candies into their respective bins, has been malfunctioning.

As a result, candies of various colors are getting mixed up, causing confusion and frustration for the workers. Adam and Laila realized that this was a problem that could be solved with colors sorting system.



Intro: Using Engaging Strategies

Strategy: Case Scenario Exploration

- **Implementation:**
 1. Display the case scenario on the board or hand out printed copies.
 2. Read the scenario aloud and highlight key problems (e.g., inefficiency, worker frustration).
 3. Ask students to imagine they are factory workers dealing with mixed-up candies.

Strategy: Hands-on Demonstration (Sorting Challenge)

- **Implementation:**
 1. Provide students with a container of mixed colored beads/buttons.
 2. Time how long it takes them to sort manually.
 3. Ask: "How could we automate this process to make it faster?"

Story Reflection

Questions:

1. Why do factories need machines to sort items by color?
2. What could happen if different colors get mixed up in a product line?

3. *How might automation help solve this issue?*

Suggested Answers:

1. Factories need sorting machines to increase speed, reduce errors, and improve efficiency.
2. Mixing colors could lead to defective products, customer complaints, and production delays.
3. Automation uses sensors and programmed systems to ensure accuracy and efficiency in sorting.

Strategy: Think-Pair-Share Discussion

- **Implementation:**

1. **Think:** Students reflect individually on the questions.
2. **Pair:** Discuss ideas with a partner.
3. **Share:** Each pair presents their solutions to the class.

Strategy: Class Brainstorming Web

- **Implementation:**

1. Write "Color Sorting Machine" in the center of the board.
2. Ask students to brainstorm ways automation could fix the problem.
3. Connect their ideas to form a solution map.

2. Think:

Questions and Possible Answers:

- **How do you think production engineers contribute to improving the economy?**
Engineers improve production efficiency, reduce costs, and enhance product quality, leading to economic growth.
- **How can we design a color sorting system that accurately separates candies by color, ensuring efficiency and reducing errors in a factory setting?**
By using sensors to detect colors, motors to move candies, and a programmed system to sort them efficiently.

Brainstorm Web:

Strategy: Brainstorm Web

- **Implementation:**

1. Draw a Brainstorm Web on the board with "*Production Engineers in Industry*" in the center.
2. Ask students: "*How do production engineers contribute to automation?*"

3. Write their responses (e.g., improving efficiency, reducing waste, lowering costs) as branches.
4. Expand the web to include *"Designing a Color Sorting System."*
5. Have students brainstorm components (e.g., sensors, motors, programming logic) that would improve sorting accuracy.
6. Connect their ideas to form a solution map.

Strategy: Think-Pair-Share Discussion

- **Implementation:**
 1. Think: Students individually reflect on the role of engineers in automation.
 2. Pair: Discuss ideas with a partner.
 3. Share: Each pair presents their solutions to the class.

3. Explore:

Teacher Role:

- Guide students to observe the color sorting process.
- Encourage them to think about how sorting can be automated in a factory.
- Ask students to discuss the challenges they face while manually sorting.
- Relate the activity to real-world sorting systems, such as color sorting machines.

Expected From Students:

- Actively participate in the hands-on sorting activity.
- Identify patterns and methods for efficient sorting.
- Discuss the importance of sorting in production and manufacturing.

Parts of the Books Included:



Activity - Do It: choose one from these activities

1- Sorting Colored Balls into Containers

Materials Needed:

- A container filled with balls of various colors.
- Several empty containers (one for each color).
- Stopwatch or timer (to measure sorting time).

Step-by-Step Instructions:

1. Place all the colored balls in one large container.
2. Ask students to sort the balls manually into separate containers based on color.
3. Time how long it takes to sort all the balls.
4. Discuss:
 - What methods did students use to sort faster?
 - What challenges did they face?
5. Introduce the concept of automation and ask:
 - How could a machine do this task more efficiently?
 - What sensors could be used in a real color-sorting system?

Reflection Questions:

1- What was the most challenging part of sorting manually?

Sorting took a long time, and sometimes mistakes happened when placing balls in the wrong containers.

2- How do you think a machine could improve this process?

A machine could sort the balls much faster and with higher accuracy, reducing human errors.

3- What kind of sensor could help in sorting colors?

A **color sensor** could detect different colors and send signals to a mechanism that places each ball in the correct container.

2- Online Color Sorting Game:

Game: *Nuts Puzzle - Sort by Color*

Play Here: [Crazy Games - Nuts Puzzle](#)

Teacher Role:

- Introduce the game as a simulation of color sorting in a factory.
- Encourage students to analyze the strategies used in the game.
- Guide a discussion on how this digital activity relates to real-world automation.

Expected From Students:

- Play the online game and observe how sorting is done logically and efficiently.
- Identify patterns and strategies used in sorting.
- Compare their manual sorting experience with the game's automated approach.

Activity - Do It: Playing the Color Sorting Game**Materials Needed:**

- Access to the game on a computer or tablet.
- A stable internet connection.

Step-by-Step Instructions:

- 1- Open the [game link](#).
- 2- Allow students 5-10 minutes to explore and play.
- 3- Ask them to observe how the game logic sorts objects.
- 4- Discuss how the game simulates real-world color sorting machines.
- 5- Compare the game to the manual sorting activity.

Reflection Questions:

- 1- **What strategy did you use to sort colors in the game?**
I grouped similar colors together and tried to complete each set before moving to the next.
- 2- **How does this game relate to real-world color sorting machines?**
The game uses **rules and logic** to automatically sort objects, just like machines do in factories using sensors and programmed instructions.
- 3- **Why is sorting important in a factory setting?**
Sorting ensures that products are properly organized, reduces mistakes, and increases production efficiency.

4. Explain:

Teacher Role:

- Explain the role of Production Engineers in manufacturing.
- Guide students in understanding how engineers optimize processes for efficiency.
- Facilitate discussions on how engineering innovations impact the economy and industry.
- Encourage students to connect the video content to the color sorting case scenario.

Expected From Students:

- Engage actively by watching the video and taking notes.
- Discuss how production engineers solve real-world manufacturing challenges.
- Identify key responsibilities of production engineers in factories.
- Reflect on how engineering impacts automation and efficiency.

Parts of the Books Included:



Watch It: Video Activity

Video Title: *What Do Production Engineers Do?*

Video Link: [Watch Here](#)

Pre-Watching: Setting a Purpose

Strategy: KWL Chart (Know-Want to Know-Learned)

- **Implementation:**
 1. Draw a **KWL Chart** on the board:
 - **K (Know):** Ask students what they already know about Production Engineers.
 - **W (Want to Know):** Have students list questions they have before watching.
 2. Tell students to watch the video carefully to find answers.

K-W-L Chart

Topic: _____

K	What I Know:
W	What I Want to Know:
L	What I Have Learned:

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During Watching: Active Engagement

Strategy: Guided Note-Taking (Engineering Focus)

- **Implementation:**

1. Provide students with a Guided Note-Taking Sheet with prompts:
 - What tasks do Production Engineers handle in factories?
 - How do they improve production efficiency?
 - What skills are important for a Production Engineer?
2. Pause the video at key moments to highlight important details.
3. Encourage students to jot down key points and observations.

Post-Watching: Discussion & Reflection

Strategy: Think-Pair-Share

- **Implementation:**

1. **Think:** Give students 2 minutes to reflect on their notes.
2. **Pair:** Have them discuss their key takeaways with a partner.
3. **Share:** Invite pairs to share their insights with the class.

Read About It: "Automation, Robotics, and the Factory of the Future"

Pre-Reading: Discussion

Teacher Role:

- Introduce the topic by asking students about their thoughts on automation and robotics in factories.
- Guide a discussion on how machines help in improving efficiency and reducing human effort in industries.

- Relate the discussion to real-world examples like automatic packaging, robotic arms in assembly lines, and AI-driven sorting machines.

Expected From Students:

- Share prior knowledge about automation in daily life (e.g., vending machines, automatic doors).
- Discuss how machines might replace or assist human workers in different industries.
- Predict what they might learn from the article.

Strategy: Think-Pair-Share

- **Implementation:** Ask students to think about industries where automation is used, discuss with a partner, and share insights with the class.

During Reading: Engagement

Teacher Role:

- Guide students to focus on key points in the article, such as the role of robotics, the benefits of automation, and its impact on factory workers.
- Encourage note-taking on important concepts like AI in manufacturing, cost efficiency, and real-world examples of automated systems.

Expected From Students:

- Read actively, highlight key ideas, and take notes.
- Identify new technologies mentioned in the article and relate them to what they have seen in videos or discussions.
- Ask questions about unclear concepts and discuss them with classmates.

Strategy: Annotation & Highlighting

- **Implementation:** Students highlight terms related to automation and write short notes explaining how each concept applies to real-world industries.

Post-Reading: Summarize and Reflect

Teacher Role:

- Facilitate a discussion on how automation and robotics impact production efficiency.
- Ask students to summarize the key takeaways from the reading.
- Relate the reading back to the **Color Sorting Machine** project, discussing how automation improves accuracy and speed in sorting.

Expected From Students:

- Summarize the main ideas from the article in their own words.

- Discuss the role of robotics and automation in their everyday lives.
- Reflect on how automation could change future careers and industries.

Strategy: Exit Ticket

- **Implementation:** Before leaving, each student writes down one key takeaway from the article and one question they still have about automation and robotics.

5. Elaborate:

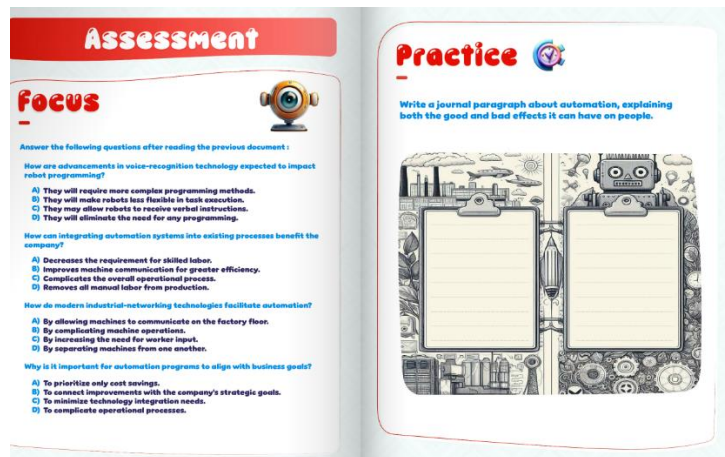
Teacher Role:

- Encourage students to apply their knowledge of automation in real-world scenarios.
- Guide students in critically analyzing the impact of automation on industries and daily life.
- Support students in articulating their thoughts through structured writing.

Expected From Students:

- Answer focus questions with reasoning and explanations.
- Participate in a writing activity to express their understanding of automation.
- Reflect on the balance between the benefits and challenges of automation.

Parts of the books included:



Assessment

Focus Questions:

1. **How are advancements in voice-recognition technology expected to impact robot programming?**
 - Correct Answer: C) They may allow robots to receive verbal instructions, making them more user-friendly.
2. **How can integrating automation systems into existing processes benefit the company?**

- Correct Answer: B) It improves machine communication for greater efficiency, optimizing production.

3. **How do modern industrial-networking technologies facilitate automation?**

- Correct Answer: A) By allowing machines to communicate on the factory floor, enabling seamless operations.

4. **Why is it important for automation programs to align with business goals?**

- Correct Answer: B) To connect improvements with the company's strategic goals, ensuring efficiency and growth.

Practice Activity: Journal Writing

Materials Needed:

- Pen and paper or digital writing tools.

Step-by-Step Instructions:

1. **Introduction:** Start by defining automation in your own words.
2. **Positive Effects:** Describe how automation benefits industries and workers.
3. **Negative Effects:** Discuss the challenges, such as job displacement or ethical concerns.
4. **Personal Reflection:** Share your thoughts on whether automation is more beneficial or harmful.

Reflection Questions:

- What industries benefit the most from automation?
- How can automation be improved to reduce negative impacts on workers?
- How do you feel about the role of automation in the future?

Challenge:

At-Home Task: Research an automation system (e.g., smart homes, self-driving cars) and write a short report on how it is changing daily life.

6. Evaluate:

Teacher Role:

- Facilitate the assessment process by reviewing students' EDP (Engineering Design Process) papers to ensure they have effectively completed the "Ask" phase.
- Provide constructive feedback on their problem identification and proposed solutions.
- Use a rubric to assess clarity, creativity, and depth of student responses.

Expected From Students:

- Answer key assessment questions related to automation and sorting systems.
- Demonstrate their understanding by completing the "Ask" phase of their EDP paper with well-thought-out responses.
- Clearly articulate the problem statement and describe how a color sorting machine can solve real-world manufacturing challenges.

Showcase: Evaluating the Main Project

Each student's EDP paper (Ask phase) should include:

- A well-defined problem statement.
- An explanation of why a color sorting machine is needed.
- Consideration of real-world constraints (cost, accuracy, speed).

Rubric for Assessment

Criteria	Excellent (4)	Good (3)	Satisfactory (2)	Needs Improvement (1)
Problem Statement	Clearly defines the problem and its real-world importance.	Defines the problem with some real-world context.	Problem is stated but lacks clarity or depth.	Problem is unclear or missing.
Understanding of Automation	Demonstrates a strong understanding of automation in sorting.	Shows good knowledge of automation principles.	Some automation concepts are present but not well developed.	Lacks understanding of automation.
Creativity in Identifying Solutions	Offers a highly creative and well-reasoned approach.	Suggests a logical solution with some creativity.	Provides a basic solution with limited creativity.	Solution is unclear or not well thought out.
Consideration of Constraints	Thoughtfully discusses real-world constraints (cost, accuracy, speed).	Mentions some constraints but lacks detail.	Constraints are minimally considered.	Constraints are missing.

Relation to Standards

- **NGSS (Next Generation Science Standards):**
 - **MS-ETS1-1:** Define the criteria and constraints of a design problem with sufficient precision to ensure a successful solution.
 - **MS-ETS1-3:** Analyze data from tests to determine similarities and differences among several design solutions.

- **Common Core Standards:**

- **CCSS.ELA-LITERACY.W.8.1:** Write arguments to support claims with clear reasons and relevant evidence.
- **CCSS.MATH.PRACTICE.MP4:** Model with mathematics (e.g., discussing constraints in automation).

Lesson 2

UNIT 01: {Beginner}

Chapter 01: Mechanics

Lesson 01: Case scenario

Time Frame: 90 minutes

Objectives:

-

Challenge Questions of the Lesson

ESL Relation

Vocabulary Development:

Introduce key terms like:

Answers to Challenge Questions

1.

Answers of Exercises

2.

1. Engage:

Color Sorting Machine

Learning Outcomes

By the end of the theme student will be able to:

- Select Optimal Cam Type.
- Identify the best belt type for your system.
- Research and apply methods to prevent jams.
- Ensure smooth object flow through the system.
- Identify Belt Drive Components: Recognize pulleys, shafts, and belts.
- Create conveyors and bins for sorting.
- Assemble System: Accurately build the sorting system.
- Create Linkages: Design linkages to control sorting arms or gates.
- Wire Color Sensor: Connect a color sensor to an electrical circuit.
- Control Multiple Servos: Connect and synchronize multiple servo motors.
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- Power and Sync Servos: Configure the circuit to power and sync servos.
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Contents

<u>Lesson 1</u>	<u>3</u>
<u>1. Engage:</u>	<u>4</u>
<u>2. Think:</u>	<u>6</u>
<u>3. Explore:</u>	<u>7</u>
<u>4. Explain:</u>	<u>10</u>
<u>5. Elaborate:</u>	<u>13</u>
<u>6. Evaluate:</u>	<u>14</u>

Lesson 2

UNIT 01: [Beginner]

Chapter 01: Mechanics

Lesson 02: Belt design

Time Frame: 90 minutes

Activity Name	Duration	Description
Engage – Conveyor Belt Memory Challenge	10 min	Students get immersed in a story about a problem in chocolate factory machines
Think – Brainstorm Web	10 min	Students create a brainstorm web to explore key ideas about color detection, belt movement, system design, and maintenance.
Explore – Conveyor Belt Demonstration	15 min	The teacher introduces different types of conveyor belt systems using real-life examples or a video demonstration. Students discuss key observations.
Explain – Understanding Belt Drive Systems & Cam	20 min	Using the provided video and book sections, students learn about belt types, pulley systems,

Mechanisms		tension management, and cam mechanisms through guided discussion.
Elaborate – Conveyor Belt Selection & Construction	25 min	Students choose the right conveyor belt for their color sorting machine, select materials, and begin designing their belt-driven system on paper.
Evaluate – EDP Paper & Reflection	10 min	Students complete the Plan section of their EDP paper, listing steps for building their conveyor system. The teacher reviews and provides feedback.

1. Engage:

Teacher Role:

- Present a relatable real-world scenario to spark curiosity.
- Facilitate a Think-Pair-Share discussion, guiding students to identify possible causes of system failures.
- Connect students' ideas to the lesson's key concepts.

Expected From Students:

- Actively **listen** to the scenario and **visualize** the problem.
- **Think individually** about possible causes of failure.
- **Discuss** their ideas with a partner and then share with the class.

Intro Story – The Misaligned Conveyor

Strategy Used: *Problem-Solving & Predicting Outcomes*

How to Implement:

1. Story Setup (Teacher Narrates)

- "Imagine you are working in a chocolate factory. A conveyor belt moves chocolates from the production area to the packaging section. Suddenly, chocolates start piling up, some falling off, and the whole system stops working."
- "Your manager says, 'We need to fix this fast! But first, we need to understand what went wrong'"

2. Think-Pair-Share Activity:

- **Think:** Individually, students brainstorm what might have caused the malfunction.
- **Pair:** Students discuss their ideas in pairs, listing possible reasons.
- **Share:** Groups share their answers with the class.

Story Reflection

Discussion Questions:

1. *What are some possible reasons why the conveyor belt stopped working?*
2. *How do you think factories prevent this kind of issue?*
3. *What role do belts, pulleys, and tension play in keeping a system running smoothly?*
4. *What could be done to fix this problem?*

Suggested Answers:

1. The belt might be loose or broken.
2. Factories maintain proper belt tension and regularly check the system.
3. Belts and pulleys work together to move objects smoothly—tension is important to keep things running.
4. Adjust the belt, check alignment, and replace broken parts.

Teaching Strategies for Reflection:

Predict & Explain: Students make educated guesses about the system failure and justify their reasoning.

Class Brainstorm Web: The teacher draws a web on the board with “Why Did the Conveyor Fail?” in the center, adding students’ ideas around it.

Mini-Demonstration: If possible, show a small belt system (toy car, fan belt, or a simple model) to reinforce concepts.

2. Think:

Guiding Questions & Suggested Answers:

1. How does our sensor detect colors?

Answer: The sensor detects colors by analyzing the light reflected from an object. It measures red, green, and blue (RGB) values and determines the color based on their intensity.

2. What factors should we consider to ensure the belt moves smoothly?

Answer: Factors include proper tension, alignment of pulleys, friction, belt material, and avoiding obstacles that may cause jams.

3. What is our plan for designing the most effective belt?

Answer: Choosing the right belt type, ensuring proper tension, aligning pulleys correctly, and considering load capacity and speed requirements.

4. How will we maintain the system?

Answer: Regularly checking for wear and tear, adjusting tension, cleaning sensors and belts, and ensuring no debris affects movement.

Teaching Strategies for This Part:

1. **Think-Pair-Share** – Students first think about the answers individually, then discuss with a partner before sharing with the class.
2. **Brainstorm Web** – Students create a visual web connecting the main challenges and solutions.
3. **Group Discussion** – Split the class into small groups, assigning each a question to discuss and present their conclusions.
4. **Concept Mapping** – Students create a flowchart showing how the belt system, sensors, and maintenance steps interconnect.

3. Explore:

Teacher Role:

- Guide students through the memory challenge by showing them the video or a real-life conveyor belt with objects.
- Encourage students to observe closely and remember as many items as possible.
- Facilitate discussion on memory strategies and how observations relate to real-world belt systems.
- Lead a reflection on how attention to detail is essential for designing and maintaining an efficient belt system.

Expected From Students:

- Watch the conveyor belt display carefully and try to memorize as many objects as possible.
- Recall and list the items within 45 seconds after the display ends.
- Discuss strategies that helped them remember objects effectively.
- Relate the activity to real-world conveyor belt functions in industries.

Parts of the Books Included:

Explore

Do it...

Try to memorise as many items from the conveyor belt as you can. Then you have 45 seconds to see how many of the items you can recall.

[Click here to play the "Guess the Number Challenge"](#)

Activity - Do It: Conveyor Belt Memory Challenge

Materials Needed:

- **Video:** ["Guess the Number Challenge"](#)
- Whiteboard or paper for students to write down recalled objects.

Step-by-Step Instructions:

1. **Setup:** Introduce the activity and explain the objective—students must observe and memorize items on the conveyor belt.
2. **Watch the Video:** Play the video once.
3. **Memory Test:** After the video ends, give students **45 seconds** to write down as many objects as they can recall.
4. **Compare & Discuss:**
 - Who remembered the most items?
 - What strategies helped in memorizing?
 - What factors made memorization difficult?
5. **Link to Engineering Concepts:** Discuss how conveyor belts are used in **sorting, automation, and efficiency**.

Reflection Questions:

1. **How does this challenge relate to how a belt system operates?**

- *(Belt systems move items in industries, requiring precision and coordination.)*
2. **What challenges did you face in recalling items? How can we apply this to real conveyor belt systems?**
- *(Proper sorting, tracking, and automation improve real conveyor belt efficiency.)*
3. **How do factories ensure smooth operation of conveyor belts?**
- *(They use sensors, maintenance checks, and automated controls to prevent jams and errors.)*

4. Explain:

Teacher Role:

- Guide students in understanding how the color sensor detects colors and how belt systems function.
- Encourage students to analyze key concepts from the video and relate them to real-world applications.
- Facilitate discussion on challenges in designing and maintaining belt systems.

Expected From Students:

- Engage in discussions by sharing observations and key takeaways from the video.
- Answer guiding questions and explain how sensors and belt mechanisms work.
- Connect the concepts to their project and suggest improvements for efficiency.

Parts of the Books Included:

Explore

Do it...

Try to memorise as many items from the conveyor belt as you can. Then you have 45 seconds to see how many of the items you can recall.

[Click here to play the "Guess the Number Challenge"](#)

Watch It: Video Analysis

□ **Video Link:** [Conveyor Belt xplanation](#)

Pre-Watching: Setting the Purpose

- **Ask:** *"What do you expect to see in this video? How do you think color sensors work?"*
- **Discuss:** The importance of automation in industries using color sensors and conveyor belts.

During Watching: Active Engagement

- Students take notes on:
How the **conveyor belt moves objects smoothly**.
What challenges may arise in the system and how they are managed.

Post-Watching: Discussion & Reflection

Questions:

1. *How does the sensor differentiate between colors?*
 - **Answer:** It detects reflected light wavelengths and assigns values to specific colors.
2. *Why is accurate color detection important in sorting systems?*
 - **Answer:** It ensures correct classification of objects in industrial automation.
3. *What problems might occur in the belt system?*
 - **Answer:** Jams, misalignment, sensor errors, and inconsistent belt speed.
4. *What are possible solutions to ensure smooth operation?*
 - **Answer:** Proper belt tension, regular calibration of sensors, and error detection algorithms.

Teaching Strategies:

- **Think-Pair-Share:** Students discuss answers before sharing with the class.
- **Concept Mapping:** Draw a flowchart of how the color sensor and belt interact.

5. Elaborate:

Teacher Role:

- Encourage creativity by guiding students in applying their knowledge to build a functional conveyor system.
- Provide feedback on their designs and challenge them to optimize their conveyor belts for efficiency.
- Ensure students connect theoretical knowledge with practical implementation.

Expected From Students:

- Engage in hands-on learning by selecting the correct conveyor type for a specific task.
- Build and test a model of a conveyor system.
- Justify their design choices and evaluate system efficiency.

Parts of the Books Included:

Choose the type of conveyor belt that best suits your machine, choose your materials and start to build it.



Assessment

Focus:

Question: *Which type of conveyor is specifically designed for transporting goods under special circumstances?*

Answer: A. Special belt conveyor

Question: *What are the key differences between a cleated belt conveyor and an inclined belt conveyor?*

Answer:

- **Cleated Belt Conveyor:** Uses raised sections (cleats) to prevent materials from sliding.
- **Inclined Belt Conveyor:** Angled for upward transport but lacks cleats.

Strategies for Answering:

- Process of Elimination: Rule out incorrect options based on belt functions.
- Compare & Contrast: Break down differences by function and structure.
- Real-Life Examples: Relate to factory setups students may have seen.

Practice

"Choose the type of conveyor belt that best suits your machine, choose your materials, and start to build it."

Materials Needed:

- Cardboard sheets
- Small motors
- Rubber bands (to simulate belts)
- Wooden sticks (for rollers)
- Hot glue and tape
- Color markers (to label belt types)

Step-by-Step Instructions:

1. **Select a Conveyor Type:** Based on function (flat, cleated, inclined).
2. **Design Your Model:** Sketch how the belt will move objects.
3. **Build the Frame:** Use cardboard and wooden sticks to create structure.
4. **Attach the Belt:** Secure rubber bands to mimic belt movement.
5. **Test & Adjust:** Ensure smooth operation and troubleshoot issues.
6. **Present & Justify:** Explain why this belt type was chosen.

Reflection Questions:

1. *Why did you choose this specific conveyor type?*
2. *How does the belt system ensure smooth operation?*
3. *What problems did you encounter, and how did you solve them?*

6. Evaluate:

Teacher Role:

- Facilitate assessments by guiding students through their EDP (Engineering Design Process) paper to ensure they outline the steps accurately.
- Provide constructive feedback on students' color sorting machine prototypes based on functionality and design.
- Observe student engagement and problem-solving strategies during the building process.
- Use the rubric to assess both conceptual understanding and practical execution.

Expected From Students:

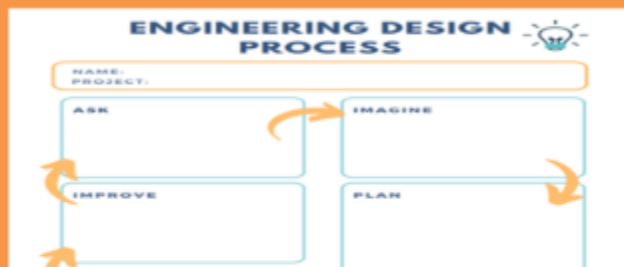
- Complete the "Plan" section of the EDP paper, listing the steps needed to build their conveyor system.

- Begin assembling their color sorting machine using proper belt drive system knowledge.
- Justify their design choices and explain how they ensured smooth belt movement.
- Reflect on challenges faced and solutions applied.

Parts of the Books Included:



After we learned what are the parts we need for our machine, let's go back to our EDP paper and add these parts to the planning part..



Add On Note



Have you heard about Cam and Follower?
Watch these videos to understand more about it:
<https://youtu.be/EXeKyTGLmzo>
<https://youtu.be/vRnA5mdia40>

Showcase

Make the body for the color sorting machine.

Showcase Activity:

Task 1: Fill in the Plan section of the EDP paper with step-by-step instructions for building their conveyor.

Task 2: Begin building the color sorting machine, ensuring that the belt moves smoothly and operates effectively.

Task 3: Present their work-in-progress and explain how their conveyor system will contribute to sorting objects based on color.

Rubric for Evaluation

Criteria	Excellent (4 pts)	Good (3 pts)	Needs Improvement (2 pts)	Incomplete (1 pt)
EDP Paper (Plan Section)	Clearly outlines all steps with details and diagrams.	Lists most steps but lacks some clarity or visuals.	Steps are unclear or missing key details.	Minimal effort, incomplete plan.
Understanding of Belt Drive System	Demonstrates a strong grasp of pulleys, belts, and system flow.	Understands the basics but has minor misconceptions.	Shows limited understanding; confused about components.	Does not demonstrate understanding.
Belt Functionality in Prototype	Belt moves smoothly and aligns properly.	Belt moves with minor issues but functions.	Belt moves inconsistently or requires manual adjustment.	Belt does not move or is improperly installed.
Use of Materials & Construction	Uses materials effectively for a strong structure.	Structure is functional but could be improved.	Some materials not used effectively; weak build.	Poor construction, materials not well utilized.
Problem-Solving & Adjustments	Identifies and fixes issues effectively.	Solves problems with some teacher guidance.	Struggles to adjust without help.	Does not attempt problem-solving.

Relation to Standards:

NGSS (Next Generation Science Standards):

- **MS-ETS1-1:** Define criteria and constraints of a design problem.
- **MS-PS2-2:** Explain how forces affect motion in mechanical systems.
- **MS-ETS1-3:** Evaluate solutions using iterative testing.

Common Core State Standards (CCSS):

- **CCSS.MATH.PRACTICE.MP4:** Model with mathematics (calculating belt tension, angles).

CCSS.ELA-LITERACY.WHST.6-8.2: Write informative/explanatory texts on technical topic

Color Sorting Machine

Learning Outcomes

By the end of the theme student will be able to:

- Select Optimal Cam Type.
- Identify the best belt type for your system.
- Research and apply methods to prevent jams.
- Ensure smooth object flow through the system.
- Identify Belt Drive Components: Recognize pulleys, shafts, and belts.
- Create conveyors and bins for sorting.
- Assemble System: Accurately build the sorting system.
- Create Linkages: Design linkages to control sorting arms or gates.
- Wire Color Sensor: Connect a color sensor to an electrical circuit.
- Control Multiple Servos: Connect and synchronize multiple servo motors.
- Calibrate Sensor: Test and fine-tune the color sensor.
- Power and Sync Servos: Configure the circuit to power and sync servos.
- Learn New Functions: Understand and apply new functions and loops.
- Link Signals: Connect sensor input to servo motor output.
- Integrate Arduino with Python: Interface Arduino with Python for control.

Contents

<u>Lesson 3</u>	<u>3</u>
<u>1. Engage:</u>	<u>4</u>
<u>2. Think:</u>	<u>5</u>
<u>3. Explain:</u>	<u>7</u>
<u>4. Explore:</u>	<u>9</u>
<u>5. Elaborate:</u>	<u>11</u>
<u>6. Evaluate:</u>	<u>14</u>

Lesson 3

UNIT 01: {Beginner}

Chapter 02: Design

Lesson 03: Do we need actuators

Time Frame: 90 minutes

Activity Name	Duration	Description
Engage: Real-World Connection	10 min	Discuss real-world applications of color sorting machines in recycling, food processing, and industry. Use videos or images to spark curiosity.
Think: Brainstorm & Design Discussion	10 min	Students brainstorm key components of a sorting system, discussing the role of actuators and sensors.
Explore: Fusion 360 & Actuator Implementation	20 min	Students refine their Fusion 360 designs , integrating actuators into their sorting mechanism.
Explain: Teacher-Guided Learning	15 min	Teacher explains actuator functionality, movement mechanics, and troubleshooting techniques for smoother sorting operations.

Elaborate: Prototype Testing & Adjustments	15 min	Students simulate or prototype their designs, refining actuator integration for efficiency.
Evaluate: Showcase & Assessment	20 min	Students present their designs, receive peer feedback, and answer focus questions. Teachers assess using the rubric.

1. Engage:

Teacher Role:

- Facilitate discussions and guide students to analyze a real-world problem.
- Ask thought-provoking questions to encourage critical thinking.
- Introduce the problem-solving challenge by connecting it to the lesson's objectives.

Expected From Students:

- Actively participate in discussions, share ideas, and brainstorm solutions.
- Analyze a malfunctioning conveyor system and predict potential failures.
- Reflect on how engineers solve similar problems in real-life factories.

Intro Story: "The Factory Crisis"

Strategy: Role-Playing & Visual Analysis

Implementation:

1. Set the Scene:

- "Imagine you are part of a team of engineers working for a major factory. One morning, the factory manager calls in a panic—something has gone wrong with the sorting system!"
- "Packages are piling up, some are falling off the conveyor, and the system is completely jammed. Your mission is to investigate and come up with a solution!"

2. Show a Video or Image:

- Display [a short video or image](#) of a malfunctioning conveyor system.
- Encourage students to observe the issues carefully.

Story Reflection

Strategy: Think-Pair-Share

Implementation:

1. Ask Students: "What do you notice about the broken conveyor system?"

- Give them 30 seconds to think individually.
- Have them pair up and discuss their observations.

- Let pairs share their ideas with the class.

2. Reflection Questions & Suggested Answers:

- What do you think caused the failure in the conveyor system?
Belt misalignment, lack of maintenance, motor failure, or faulty sensors
- How do you think mechanical linkages and actuators could fix the issue?
Linkages help control movement, and actuators automate the sorting process.
- Why do factories use automated sorting systems instead of manual labor?
Automation is faster, reduces errors, and increases efficiency.

3. Transition to Lesson Concepts:

- *"Now that we've analyzed the problem, let's learn how to design and control mechanical linkages using CAD software to create a fully functional sorting system!"*

2. Think:

Questions and Possible Answers:

1. What factors should be considered when designing mechanical components to ensure they work seamlessly together in a complex system?

- Proper alignment to avoid friction and inefficiency.
- Selection of appropriate materials for durability and performance.
- Reduction of friction to ensure smooth operation.
- Ensuring durability for long-term functionality.
- Choosing a suitable power source for efficiency.
- Integrating components properly to avoid mechanical failures.

2. How does the choice of actuator impact the performance and functionality of a mechanical system?

- Electric actuators provide precise control and moderate speed, making them ideal for automation.
- Hydraulic actuators generate high force but move slower, making them suitable for heavy-duty tasks.
- The right actuator ensures efficiency, power management, and accuracy in mechanical operations.

Brainstorm Web:

Central Topic: Designing a Functional Sorting System

Key Branches:

1. Mechanical Components

- Gears
- Linkages

- Belts
- Frames

2. Factors for Seamless Operation

- Alignment & Calibration
- Material Strength
- Speed & Efficiency
- Power Supply

3. Actuator Selection

- Electric Actuators → Precise control, moderate speed
- Hydraulic Actuators → High force, slower movement

4. System Integration

- Sensors for automation
- Software (CAD) for design
- Safety mechanisms

Strategies for Delivery:

Strategy 1: Concept Mapping

- **Implementation:**
 1. Provide students with a blank brainstorm web.
 2. Ask them to write down factors that affect mechanical component performance.
 3. Have them connect their ideas with arrows, showing relationships.

Strategy 2: Think-Pair-Share

- **Implementation:**
 1. Think: Give students one minute to write their thoughts individually.
 2. Pair: Have students discuss their ideas in small groups.
 3. Share: Each group presents one key idea to the class.

Strategy 3: Real-Life Application Discussion

- **Implementation:**
 1. Show images or videos of real sorting systems.
 2. Ask: *“What choices do you think engineers made when designing this?”*
 3. Encourage students to predict how different actuators would change the system.

3. Explain:

Teacher Role:

- Guide discussions on the role of actuators in mechanical systems.
- Encourage students to analyze video content and reading material.
- Facilitate connections between concepts and real-world applications.
- Ensure active student participation through discussions and questioning.

Expected From Students:

- Engage actively by watching the video and reading the article.
- Take notes on key points during the activities.
- Share observations and ask questions.
- Participate in discussions to reflect on what they learned.

Parts of the Books Included:

Explore

Watch it...

Let's have a look on actuators and how it works in system.

<https://youtu.be/-8ztN2WCcMY>

Read it...

To learn more about actuators let's read this article.

<https://www.progressiveautomations.com/pages/actuators>

Watch It: Understanding Actuators in Action

Video Link: [How Actuators Work](#)

Pre-Watching:

- Set a Purpose: Ask students: *"How do you think machines move parts automatically?"*

- **Prediction Activity:** Have students predict what an actuator does before watching.

During Watching:

- **Active Engagement:** Ask students to take notes on:
 - The function of actuators.
 - Types of actuators (electric, hydraulic, pneumatic).
 - Real-world applications of actuators.

Post-Watching:

- **Discussion Questions:**
 1. *What are actuators, and how do they work?*
 2. *What are the differences between electric, pneumatic, and hydraulic actuators?*
 3. *Where do we see actuators in daily life?*
 4. *How can actuators be used in a color sorting machine?*
- **Suggested Strategy: Think-Pair-Share**
 - **Think:** Students reflect on their answers.
 - **Pair:** Discuss with a partner.
 - **Share:** Share insights with the class.

Read About It: Deep Dive into Actuators

Reading Link: [Progressive Automations – Actuators](#)

Pre-Reading:

- **Discussion Starter:** Ask: *“What do you already know about actuators?”*
- **Guiding Questions:**
 1. *Why do different machines require different types of actuators?*
 2. *What factors determine which actuator to use in a system?*

During Reading:

- **Engagement Strategies:**
 - **Highlighting Key Information:** Students underline main points.
 - **Comparison Chart:** Students list different actuator types and their uses.

Post-Reading:

- **Summarize and Reflect:**
 - **Quick Summary:** Each student shares one key takeaway.

- Reflection Questions:
 1. *Which actuator would be best for a conveyor belt? Why?*
 2. *How do actuators improve automation?*
- **Suggested Strategy: Exit Ticket**
 - Students write down:
 1. One thing they learned.
 2. One question they still have.

4. Explore:

Teacher Role:

- Guide students in using Fusion 360 to design candy containers based on color sorting.
- Facilitate discussions on assembling mechanical parts like conveyors and bins.
- Ask guiding questions to help students think about actuator integration in movement control.
- Monitor student progress and provide technical support.

Expected From Students:

- Use Fusion 360 to create candy container designs for color sorting.
- Apply knowledge of mechanical assembly, including conveyors and bins.
- Analyze and reflect on actuator integration for controlled movement.
- Collaborate and troubleshoot challenges during the design process.

Parts of the Books Included:

Do it...

- using fusion 360 create a design for the candy containers depending on the color. Assembling mechanical parts such as conveyors, and bins.
- Think about: **how to integrate actuators into mechanical designs to control movement?**

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Activity - Do It:

Materials Needed:

- Computer with Fusion 360 installed
- Reference images of candy sorting systems
- Notes on actuators and movement control
- If you need you can use this link: https://www.youtube.com/watch?v=oeOC50-QBeU&ab_channel=MakeandRemake

Step-by-Step Instructions:

1. Brainstorm & Plan:

- Discuss how actuators control movement in a sorting system.
- Sketch initial candy container designs based on color sorting.

2. Fusion 360 Design:

- Open Fusion 360 and create a new project.
- Design candy containers that align with conveyor movement.
- Integrate compartments to sort candies based on colors.

3. Assembly Considerations:

- Add hinges or sliding mechanisms for actuator movement.
- Ensure structural stability for effective sorting.

4. Final Touches & Export:

- Review the design and adjust dimensions if needed.
- Export and save the file for future simulations or 3D printing.

Reflection Questions:

- How did you ensure the candy containers were designed for smooth sorting?
- What role do actuators play in controlling movement?
- What challenges did you face, and how did you overcome them?

5. Elaborate:

Teacher Role:

- Encourage students to apply mechanical design concepts in their prototypes.
- Guide students in identifying key components of actuators.
- Provide feedback on students' designs and how they integrate actuators into their systems.
- Foster creativity and critical thinking in problem-solving.

Expected From Students:

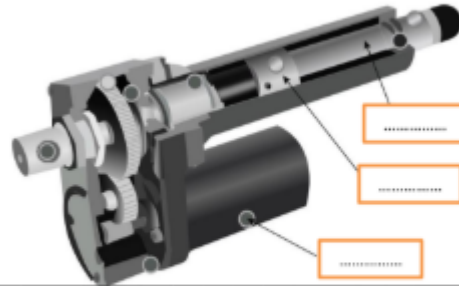
- Identify and label key actuator parts in the assessment.
- Apply actuator knowledge to enhance their prototype designs.
- Update their EDP (Engineering Design Process) paper with new components.
- Reflect on how actuator integration improves mechanical system performance.

Parts of the Books Included:

Assessment Exercise

Focus

Fill in the gap

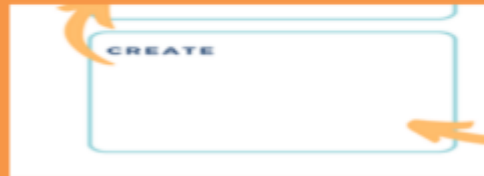


Practice

Add the design to your prototype.



After we learned what are the parts we need for our machine, let's go back to our EDP paper and add these parts to the create part.



Assessment

Focus: Fill in the Gaps

(Answers for the labeled actuator parts)

1. **Motor** – Powers the actuator to generate movement.
2. **Gear System** – Transmits and modifies motor torque.
3. **Lead Screw** – Converts rotational motion into linear motion.

Strategies for Answering:

- **Think-Pair-Share:** Have students discuss their answers before sharing them with the class.
- **Label & Explain:** Give students printed diagrams and ask them to label and describe each part.

- **Matching Activity:** Provide definitions of parts and let students match them to the correct labels.

Practice: Add the Design to Your Prototype

Materials Needed:

- Fusion 360 or paper for sketching
- Actuator component models or 3D-printed parts
- EDP paper
- Prototyping materials (cardboard, plastic, or 3D-printed parts)

Step-by-Step Instructions:

1. Review the Actuator Parts:

- Discuss the function of motors, gears, and lead screws.
- Compare different actuator types and their applications.

2. Update the EDP Paper:

- Identify the necessary actuator parts for the prototype.
- Add these parts to the "Create" section of the EDP paper.

3. Integrate the Actuator into the Prototype:

- Use Fusion 360 or physical models to assemble a working system.
- Ensure the actuator moves as expected in the prototype.

4. Test & Evaluate:

- Adjust the design if necessary to improve movement control.
- Document any challenges faced and how they were resolved.

Reflection Questions:

- How does the actuator improve your prototype's functionality?
- What challenges did you encounter while integrating it?
- How does adding actuators enhance automation in mechanical systems?

Challenge (At-Home Task):

"Design a Mini Actuator System"

- Using household materials, create a simple linear actuator system (e.g., a moving arm using a screw and a motor).
- Document the process with sketches, photos, or a short video.
- Explain how the actuator converts rotational motion into linear motion.

6. Evaluate:

Teacher Role:

- Facilitate assessments and provide constructive feedback on student projects.
- Observe students' problem-solving strategies and application of mechanical design concepts.
- Ensure students articulate their design choices and actuator integration effectively.
- Use the rubric to assess student projects based on technical accuracy, creativity, and functionality.

Expected From Students:

- Demonstrate their understanding by presenting their Fusion 360 designs and explaining their sorting system.
- Answer focus questions and reflect on challenges and solutions in the design process.
- Improve their projects based on feedback from teachers and peers.
- Document their learning process through an Engineering Design Process (EDP) paper or a short presentation.

Rubric: Color Sorting Machine Project Assessment

Criteria	Excellent (4)	Good (3)	Developing (2)	Needs Improvement (1)
Mechanical Design	Well-structured and functional, with properly aligned mechanical components.	Functional but requires slight adjustments.	Some components misaligned or need improvement.	Design lacks structure or proper functionality.
Actuator Integration	Actuators correctly implemented and fully control movement.	Actuators work but could be optimized for efficiency.	Some actuator components function, but others need adjustment.	Actuator integration is incorrect or missing.
Use of Fusion 360	Demonstrates advanced use of Fusion 360 tools and techniques.	Good use of Fusion 360, with minor issues in design.	Basic design completed, but missing key features.	Minimal effort or lack of understanding of Fusion 360 tools.
Presentation & Explanation	Clearly explains the design, actuator choices, and improvements.	Explains most design aspects with minor gaps.	Partial explanation, lacking clarity in key areas.	Little to no explanation of the design process.
Creativity & Innovation	Unique and innovative design with excellent problem-solving.	Some creative elements included.	Basic design with minimal innovation.	Lacks creativity and problem-solving elements.

Relation to Standards:

This lesson aligns with the following educational standards:

- **Next Generation Science Standards (NGSS)**
 - **MS-ETS1-1:** Define the criteria and constraints of a design problem to ensure a successful solution.
 - **MS-ETS1-3:** Analyze data from tests to determine similarities and differences among solutions.
- **Common Core State Standards (CCSS) for Mathematics**
 - **CCSS.MATH.CONTENT.7.G.A.2:** Draw, construct, and describe geometrical figures and describe relationships between them.
- **International Society for Technology in Education (ISTE) Standards**
 - **ISTE 4:** Innovative Designer – Students use a variety of technologies to create new, useful, or imaginative solutions.

Color Sorting Machine

Learning Outcomes

By the end of the theme student will be able to:

- Select Optimal Cam Type.
- Identify the best belt type for your system.
- Research and apply methods to prevent jams.
- Ensure smooth object flow through the system.
- Identify Belt Drive Components: Recognize pulleys, shafts, and belts.
- Create conveyors and bins for sorting.
- Assemble System: Accurately build the sorting system.
- Create Linkages: Design linkages to control sorting arms or gates.
- Wire Color Sensor: Connect a color sensor to an electrical circuit.
- Control Multiple Servos: Connect and synchronize multiple servo motors.
- Calibrate Sensor: Test and fine-tune the color sensor.
- Power and Sync Servos: Configure the circuit to power and sync servos.
- Learn New Functions: Understand and apply new functions and loops.
- Link Signals: Connect sensor input to servo motor output.
- Integrate Arduino with Python: Interface Arduino with Python for control.

Contents

<u>Lesson 4</u>	<u>3</u>
<u>1. Engage:</u>	<u>4</u>
<u>2. Think:</u>	<u>5</u>
<u>3. Explore:</u>	<u>6</u>
<u>4. Explain:</u>	<u>8</u>
<u>5. Elaborate:</u>	<u>10</u>
<u>6. Evaluate:</u>	<u>14</u>

Lesson 4

UNIT 02: [Intermediate]

Chapter 01: Robotics

Lesson 01: Function your system

Time Frame: 90 minutes

Activity Name	Duration	Description
1. Engage: Automation in Real Life	10 min	The teacher introduces automation and sorting systems used in industries (e.g., recycling, food packaging). Students brainstorm where sorting machines are used.
2. Think: How Does Color Sorting Work?	10 min	Students discuss how sensors detect colors and how automation classifies objects. The teacher uses Think-Pair-Share to encourage discussion.
3. Explore: Hands-on with TCS3200 Sensor	15 min	Students experiment with the TCS3200 color sensor , observe how it detects different colors, and discuss how it can be used in sorting.
4. Explain: Circuit & Code Implementation	20 min	The teacher explains how to connect the TCS3200 sensor and servo motor to Arduino and how to program the system to sort colors. Students analyze

		the code and predict its output.
5. Elaborate: Building the Color Sorting Machine	20 min	Students build and program their own color sorting system. They test the circuit, troubleshoot errors, and refine their code.
6. Evaluate: Presenting & Assessing Projects	15 min	Students present their working Color Sorting Machine , explain its function, and reflect on its real-world applications. The teacher evaluates using a rubric and provides feedback.

1. Engage:

Teacher Role:

- Facilitate a discussion about automation in factories and real-world sorting systems.
- Introduce the color sensor and servo motor as key components of automated sorting machines.
- Ask guiding questions to encourage critical thinking and problem-solving.

Expected From Students:

- Actively participate in discussions about how machines sort objects.
- Brainstorm challenges in factory automation if sensors or motors fail.
- Predict how color sensors and servo motors work in a real-world setting.
- Engage in peer discussions to share ideas and reflections.

Intro Story: "The Chocolate Factory Sorting Challenge"

Strategy: Storytelling & Real-World Connection

Implementation:

- The teacher starts with an exciting story:
"Imagine you are an engineer designing an automated sorting system for a chocolate factory. The factory produces chocolates wrapped in three colors – red, blue, and green. Your job is to create a machine that sorts them accurately and quickly using a color sensor and servo motors. But what if the system fails to detect colors correctly or the servo motors don't respond?"

Story Reflection:

Questions & Suggested Answers

- What happens if a sorting machine cannot correctly detect colors?
The factory may mix up products, leading to mistakes and wasted production
- How do you think a color sensor works to detect different colors?
It reads the reflection of light and assigns values to different colors

Strategies for Story Reflection:

Think-Pair-Share (Collaborative Thinking)

- Students think individually about the questions.

- They discuss their ideas with a partner.
- Pairs share their answers with the class.

Role-Playing (*Active Engagement*)

- A few students pretend to be color sensors (detect colors), and others act as servo motors (move objects).
- This demonstrates how sensors and motors work together before building the actual circuit.

2. Think:

Questions and Possible Answers

- 1. What could have gone wrong with the electrical system that might cause the machine to fail?**
 - Loose or disconnected wires.
 - Faulty color sensor.
 - Damaged servo motor.
 - Power supply issues.
 - Incorrect coding or logic errors in the program.
- 2. How would you troubleshoot and fix the electrical problem to get the sorter back up and running?**
 - Check all wire connections to ensure they are secure.
 - Test the color sensor separately to confirm if it is working.
 - Check the servo motor to see if it moves correctly.
 - Verify that the power supply is providing the correct voltage.
 - Debug and test the code for any errors.
 - Replace any faulty components if necessary.

Brainstorm Web: "Troubleshooting the Sorting Machine"

Implementation:

- The teacher draws a central circle on the board labeled "Sorting Machine Malfunction".
- Students contribute possible causes of failure (e.g., power issues, sensor errors, motor failure).
- The teacher adds branches for each category (hardware issues, coding errors, power problems).

- Students brainstorm possible solutions and connect them to the problem areas.

Strategies for Delivering This Part

Think-Pair-Share (*Collaborative Thinking*)

- **Think:** Students individually list possible failures and solutions.
- **Pair:** They discuss ideas with a partner.
- **Share:** Each pair presents their ideas to the class.

Problem-Solving Challenge (*Hands-on Approach*)

- The teacher provides a faulty circuit or code snippet simulating a real failure.
- Students work in small groups to diagnose and fix the problem.

Role-Playing: Troubleshooting Engineers (*Active Engagement*)

- One student acts as a machine operator explaining the problem.
- Another plays a troubleshooting engineer suggesting possible fixes.
- The class votes on the best solution before implementing it in the circuit.

3. Explore:

Teacher Role:

- Guide students in collecting the necessary components.
- Demonstrate the correct way to connect each component.
- Supervise students as they assemble the circuit and provide troubleshooting support.
- Ask students questions to reinforce understanding of each component's function.

Expected From Students:

- Collect all required components from the lab.
- Correctly identify and name each component.
- Follow instructions to assemble the color sensor circuit.
- Work in small groups to complete the activity.
- Test the circuit and troubleshoot if necessary.

Parts of the Books Included:



Activity – Do It!

Materials Needed:

- TCS3200 Color Sensor Module
- Arduino Uno Board
- Jumper Wires
- Breadboard (optional)
- USB Cable
- Computer with Arduino IDE installed

Step-by-Step Instructions:

1. **Identify and collect** all required components.

Reflection Questions:

1. What is the function of the TCS3200 color sensor in this circuit?
2. How does the Arduino receive data from the sensor?
3. What challenges did you face while wiring the circuit?
4. How can this sensor be used in real-world applications?

4. Explain:

Teacher Role:

- Explain how the color sensor works and how it interacts with the Arduino.
- Guide discussions on real-world applications of color sensors.
- Support students in analyzing the video and reading material.
- Encourage students to ask questions and share insights.

Expected From Students:

- Engage actively in discussions and ask questions.
- Observe key points while watching the video.
- Read and summarize the given material.
- Relate the concepts to practical applications.

Parts of the Books Included:



Watch It – Video Activity

Pre-Watching: Setting a Purpose

- **Strategy: KWL Chart (Know, Want to Know, Learned)**
 - Before watching, students fill in what they already *Know* about color sensors and what they *Want to Know*.
 - After watching, they complete the *Learned* section.

During Watching: Active Engagement

- **Strategy: Guided Notes**
 - Provide students with key questions to answer while watching.
 - Example questions:
 1. What does the color sensor measure?
 2. How is it connected to the Arduino?
 3. What real-life applications use this technology?

Post-Watching: Discussion and Reflection

- **Strategy: Think-Pair-Share**
 - Students first reflect on what they learned individually.
 - They then discuss with a partner before sharing with the class.
 - Teacher prompts:
 - "How does the color sensor detect different colors?"
 - "What challenges might arise when using this sensor?"
 - "Can you think of any innovative applications?"

Read About It – Reading Activity

Pre-Reading: Discussion

- **Strategy: Anticipation Guide**
 - Present statements about color sensors (true or false).
 - Example: *"Color sensors can only detect primary colors."*
 - Students predict whether the statements are true before reading.

During Reading: Engagement

- **Strategy: Annotate the Text**
 - Ask students to highlight key points and write short reflections in the margins.
 - Encourage them to note new vocabulary and summarize paragraphs.

Post-Reading: Summarize and Reflect

- **Strategy: One-Sentence Summary**
 - Students summarize the reading material in a single sentence.
 - Example: *"The TCS3200 color sensor detects color by measuring reflected light intensity and sends data to the Arduino for processing."*

5. Elaborate:

Teacher Role:

- Encourage students to experiment with the color sensor and apply what they've learned.
- Guide them in recording RGB values accurately.
- Foster creativity by allowing students to explore different color detection applications.
- Provide support and feedback on their experiments and connections.

Expected From Students:

- Construct a working color sensor circuit.
- Collect and analyze RGB data for different colors.
- Document and explain their findings.
- Think critically about how color detection is used in real-world applications.

Parts of the Books Included: